Technical Report: Healthcare Data Ingestion and Analysis using Amazon S3 and Amazon Athena

1. Objective

To document the process of building a cloud-based storage and query environment for healthcare data, from raw data ingestion to data extraction and analysis using SQL queries via Amazon Web Services (AWS).

2. Technologies & Services Used

- **Amazon S3 (Simple Storage Service):** Used for storing the raw data files and saving the generated query results.
- **Amazon Athena:** Used for creating databases and tables, and executing SQL queries directly on the data stored in S3 without the need to provision or manage servers.

3. Step-by-Step Execution Timeline

Phase 1: Data Storage Setup

1. **Bucket Creation:**
    - **Type:** General purpose.
    - **Namespace:** Account Regional namespace.
    - **Final Bucket Name:** healthcare-136876227305-us-east-1-an.
    - **Region:** us-east-1 (US East - N. Virginia).
2. **Data Ingestion:**
    - Uploaded a CSV data file named RealHealthcareData1.csv.
    - **File Size:** 1.1 KB.
    - **Storage Class:** Standard.

Phase 2: Data Cataloging (Database & Table Setup)

Using the Amazon Athena interface, the stored data was mapped to a queryable database through the following steps:

1. **Database:** Created a new database named healthcaredata.
2. **Table:** Created a table named patient_data.

3. **Dataset Location:** Pointed the table to read data from the S3 path: s3://healthcare-136876227305-us-east-1-an.
4. **Data Format:**
    - **Table type:** Apache Hive.
    - **File format:** CSV.
    - **SerDe library:** org.apache.hadoop.hive.serde2.lazy.LazySimpleSerDe.
    - **Field delimiter:** Comma ,.
5. **Schema Definition (Columns):** Defined 6 main columns:
    - Patient_ID (Data Type: int)
    - Name (Data Type: string)
    - Age (Data Type: int)
    - Diagnosis (Data Type: string)
    - Treatment (Data Type: string)
    - Files (Data Type: int)

Phase 3: Workgroup Configuration

Before executing queries, the primary workgroup settings were configured to ensure query results are managed and saved correctly:

- **Management of query results:** Selected "Athena managed" with default AWS-owned key encryption.
- **Location of query result:** Specified the exact same S3 bucket to store the output: s3://healthcare-136876227305-us-east-1-an.

Phase 4: Data Querying & Verification

To verify data integration and integrity, an SQL query was executed within the Athena Query Editor:

- **Executed Query:**

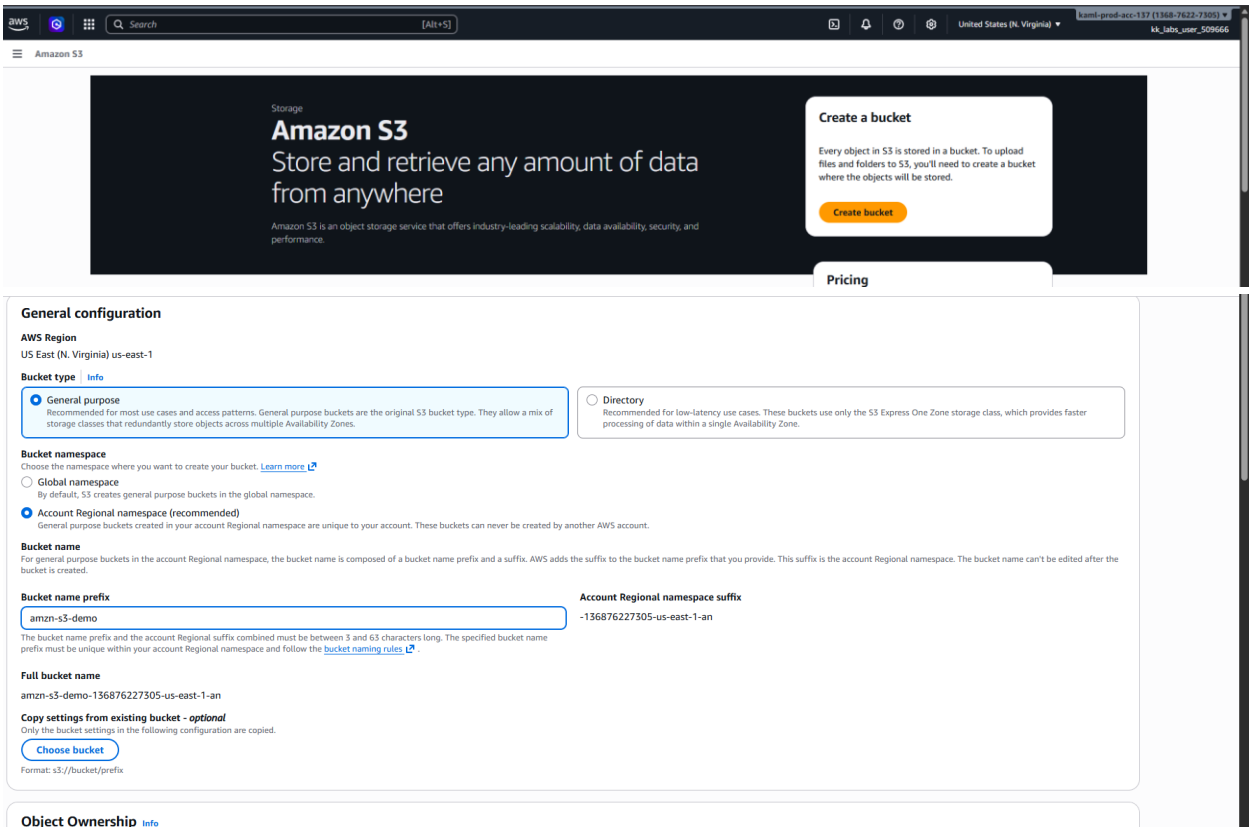
SQL

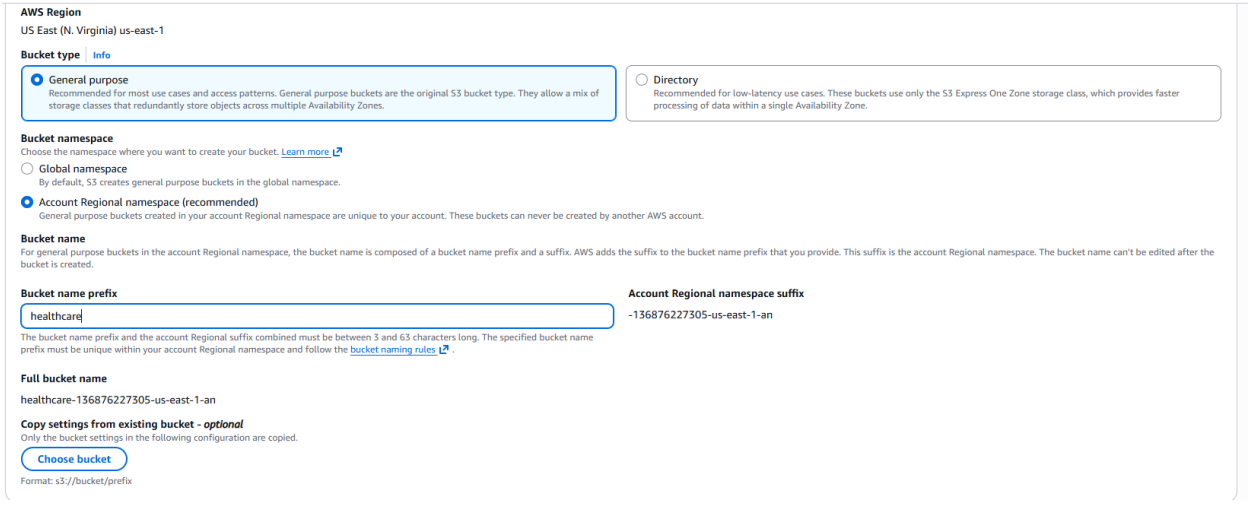
```
SELECT * FROM "healthcaredata"."patient_data" limit 10;
```

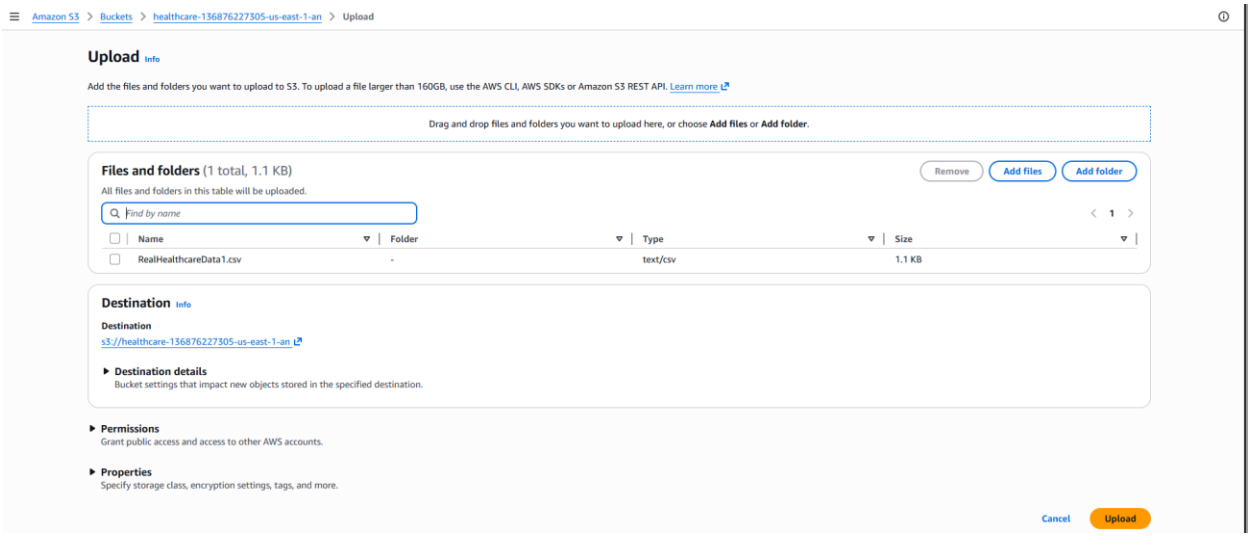
- **Execution Results:** The query completed successfully with a run time of 683 milliseconds, scanning 1.08 KB of data.
- **Output Analysis:** The results returned 10 rows of data.
    - *Technical Note:* The first row in the results displayed the original CSV text headers (e.g., "Diagnosis", "Treatment") as actual data values. This indicates that the current SerDe properties were not set to skip the first row (header row) of the source file. Subsequent records contained actual patient data (e.g., Thomas Clark, Emma Nelson) with medical diagnoses like Diabetes and Asthma.

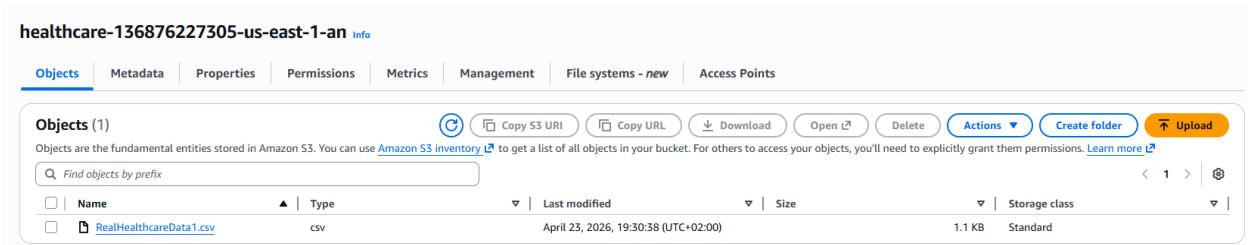
4. Conclusion

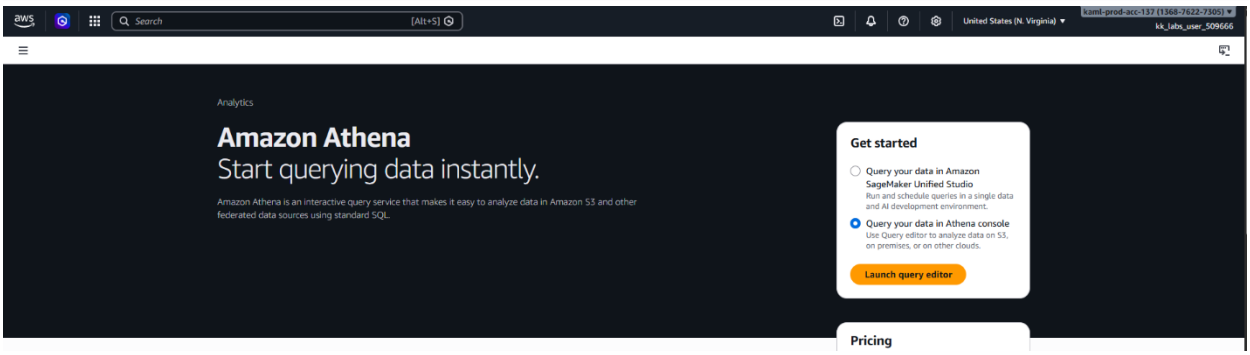
Successfully built a streamlined, serverless data pipeline on AWS. The system is now fully capable of ingesting raw data files into S3 and enabling rapid, programmatic, or standard SQL-based analysis through Athena.

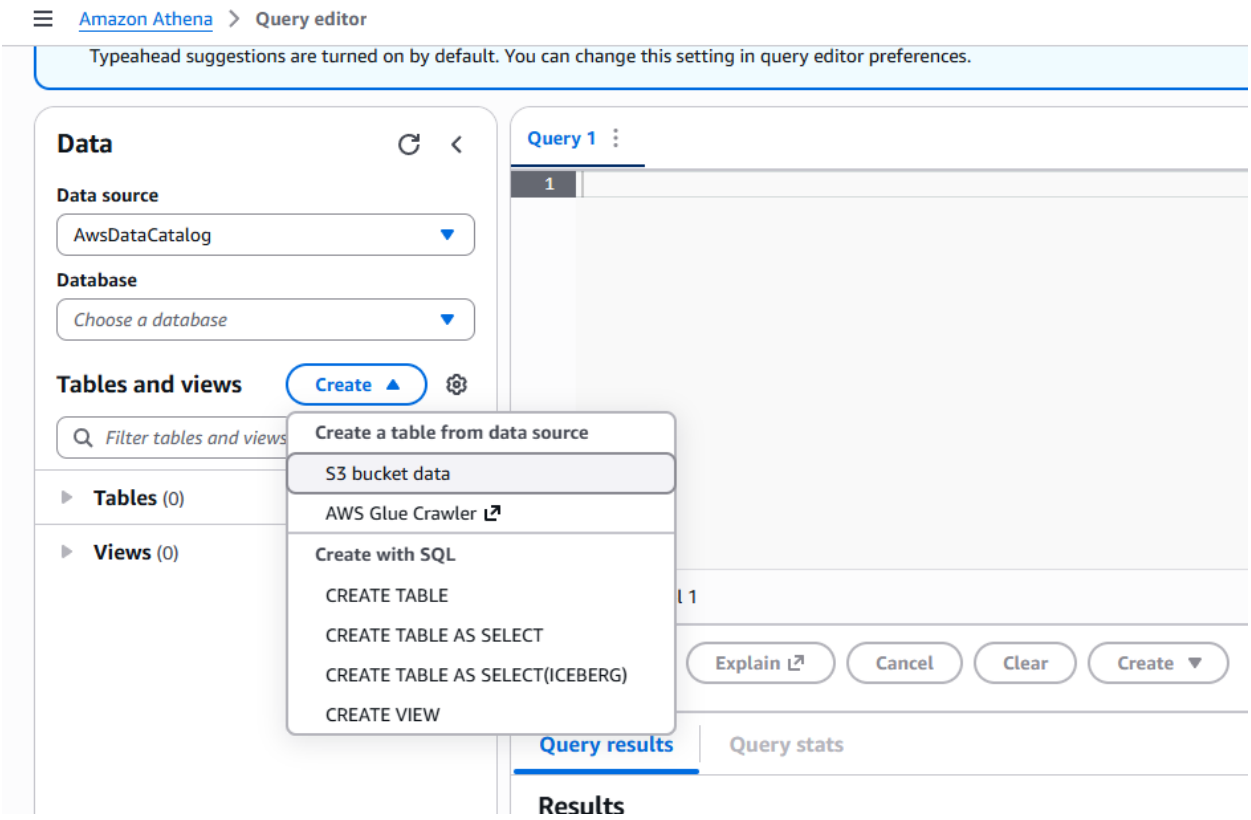


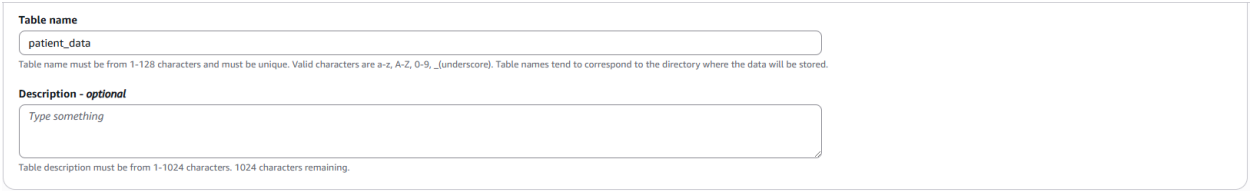

AWS Region
US East (N. Virginia) us-east-1
Bucket type Info
General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
Directory
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.
Bucket namespace
Choose the namespace where you want to create your bucket. Learn more
Global namespace
By default, S3 creates general purpose buckets in the global namespace.
Account Regional namespace (recommended)
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.
Bucket name
For general purpose buckets in the account Regional namespace, the bucket name is composed of a bucket name prefix and a suffix. AWS adds the suffix to the bucket name prefix that you provide. This suffix is the account Regional namespace. The bucket name can't be edited after the bucket is created.
Bucket name prefix
healthcare
Account Regional namespace suffix
-136876227305-us-east-1-an
The bucket name prefix and the account Regional suffix combined must be between 3 and 63 characters long. The specified bucket name prefix must be unique within your account Regional namespace and follow the bucket naming rules .
Full bucket name
healthcare-136876227305-us-east-1-an
Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.
Choose bucket
Format: s3://bucket/prefix

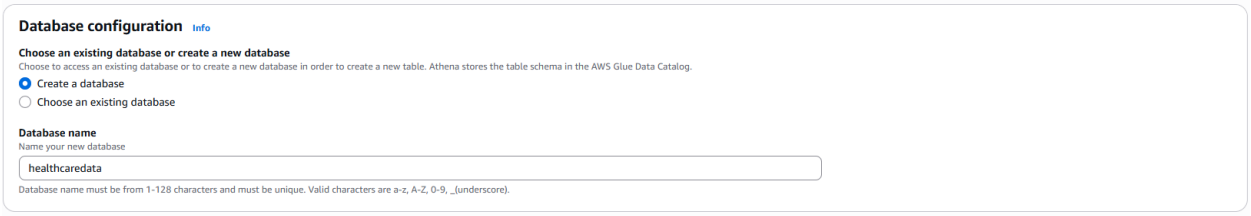

Amazon S3 > Buckets > healthcare-136876227305-us-east-1-an > Upload
Upload Info
Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. Learn more
Drag and drop files and folders you want to upload here, or choose Add files or Add folder.
Files and folders (1 total, 1.1 KB)
All files and folders in this table will be uploaded.
Remove
Add files
Add folder
Find by name
Name
Folder
Type
Size
RealHealthcareData1.csv
-
text/csv
1.1 KB
Destination Info
Destination
s3://healthcare-136876227305-us-east-1-an
Destination details
Bucket settings that impact new objects stored in the specified destination.
Permissions
Grant public access and access to other AWS accounts.
Properties
Specify storage class, encryption settings, tags, and more.
Cancel
Upload

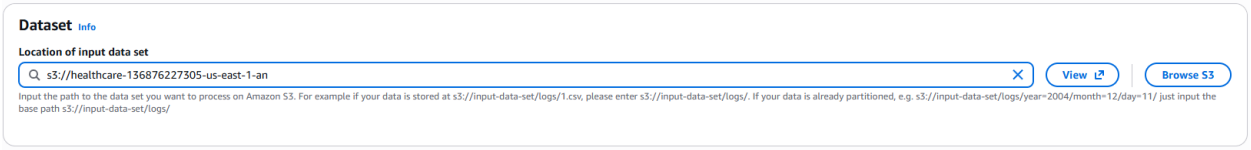

healthcare-136876227305-us-east-1-an Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems - new
Access Points
Objects (1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
RealHealthcareData1.csv
csv
April 23, 2026, 19:30:38 (UTC+02:00)
1.1 KB
Standard


Search
[Alt+S]
United States (N. Virginia)
Analytics
Amazon Athena
Start querying data instantly.
Amazon Athena is an interactive query service that makes it easy to analyze data in Amazon S3 and other federated data sources using standard SQL.
Get started
Query your data in Amazon SageMaker Unified Studio
Run and schedule queries in a single data and AI development environment.
Query your data in Athena console
Use Query editor to analyze data on S3, on premises, or on other clouds.
Launch query editor
Pricing


Amazon Athena
Query editor
Typeahead suggestions are turned on by default. You can change this setting in query editor preferences.
Data
Data source
AwsDataCatalog
Database
Choose a database
Tables and views
Create
Filter tables and views
Tables (0)
Views (0)
Create a table from data source
S3 bucket data
AWS Glue Crawler
Create with SQL
CREATE TABLE
CREATE TABLE AS SELECT
CREATE TABLE AS SELECT(ICEBERG)
CREATE VIEW
Query 1
1
Explain
Cancel
Clear
Create
Query results
Query stats
Results


Table name
patient_data
Table name must be from 1-128 characters and must be unique. Valid characters are a-z, A-Z, 0-9, _(underscore). Table names tend to correspond to the directory where the data will be stored.
Description - optional
Type something
Table description must be from 1-1024 characters. 1024 characters remaining.


Database configuration
Info
Choose an existing database or create a new database
Choose to access an existing database or to create a new database in order to create a new table. Athena stores the table schema in the AWS Glue Data Catalog.
Create a database
Choose an existing database
Database name
Name your new database
healthcaredata
Database name must be from 1-128 characters and must be unique. Valid characters are a-z, A-Z, 0-9, _(underscore).


Dataset
Info
Location of input data set
s3://healthcare-136876227305-us-east-1-an
View
Browse S3
Input the path to the data set you want to process on Amazon S3. For example if your data is stored at s3://input-data-set/logs/1.csv, please enter s3://input-data-set/logs/. If your data is already partitioned, e.g. s3://input-data-set/logs/year=2004/month=12/day=11/ just input the base path s3://input-data-set/logs/

Data format Info

Table type

Apache Hive

File format

CSV

SerDe library

org.apache.hadoop.hive.serde2.lazy.LazySimpleSerDe

SerDe properties - *optional*

Name	Value	
field.delim	,	Remove

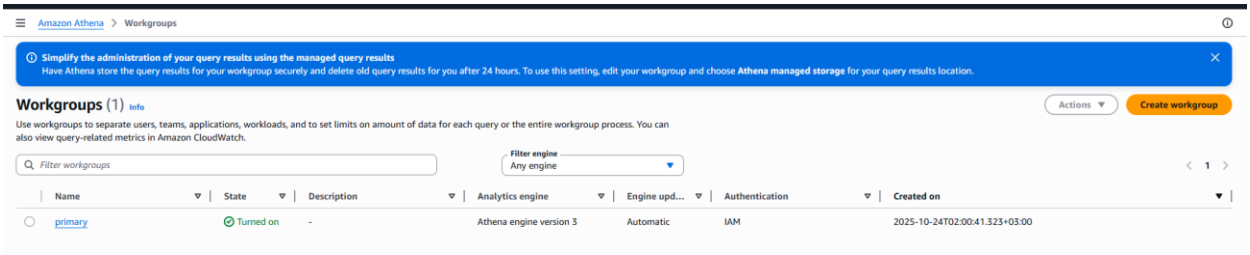
Add SerDe property

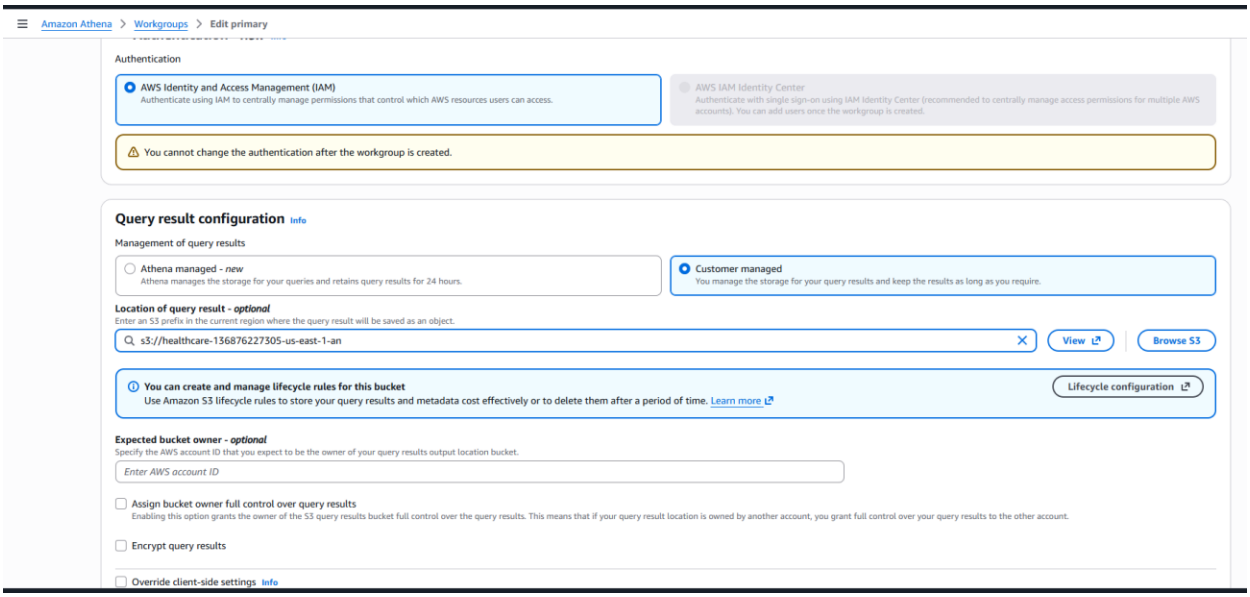
Column details

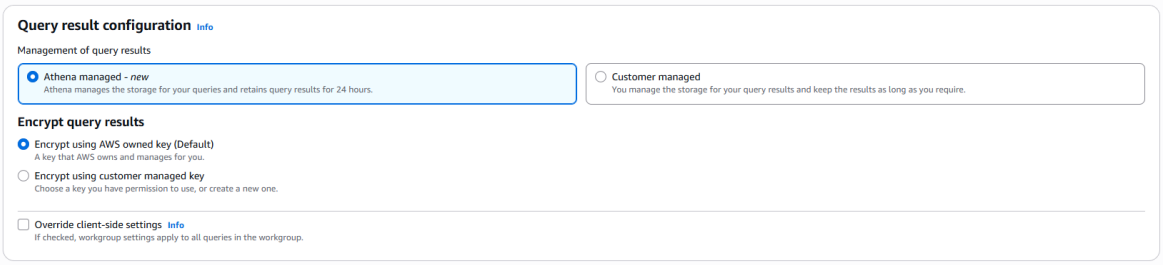
Column name must be from 1-128 characters. Valid characters are a-z, A-Z, 0-9, _(underscore). Certain advanced column types (namely, structs) are not exposed in this interface.

Column name	Column type	Description - *optional*	
Patient_ID	int	Enter description	Remove
Name	string	Enter description	Remove
Age	int	Enter description	Remove
Diagnosis	string	Enter description	Remove
Treatment	string	Enter description	Remove
Files	int	Enter description	Remove

Add a column | Bulk add columns




Amazon Athena > Workgroups > Edit primary
Authentication
AWS Identity and Access Management (IAM)
Authenticate using IAM to centrally manage permissions that control which AWS resources users can access.
AWS IAM Identity Center
Authenticate with single sign-on using IAM Identity Center (recommended to centrally manage access permissions for multiple AWS accounts). You can add users once the workgroup is created.
You cannot change the authentication after the workgroup is created.
Query result configuration Info
Management of query results
Athena managed - new
Athena manages the storage for your queries and retains query results for 24 hours.
Customer managed
You manage the storage for your query results and keep the results as long as you require.
Location of query result - optional
Enter an S3 prefix in the current region where the query result will be saved as an object.
s3://healthcare-136876227305-us-east-1-an
View
Browse S3
You can create and manage lifecycle rules for this bucket
Use Amazon S3 lifecycle rules to store your query results and metadata cost effectively or to delete them after a period of time. Learn more
Lifecycle configuration
Expected bucket owner - optional
Specify the AWS account ID that you expect to be the owner of your query results output location bucket.
Enter AWS account ID
Assign bucket owner full control over query results
Enabling this option grants the owner of the S3 query results bucket full control over the query results. This means that if your query result location is owned by another account, you grant full control over your query results to the other account.
Encrypt query results
Override client-side settings Info


Query result configuration Info
Management of query results
Athena managed - new
Athena manages the storage for your queries and retains query results for 24 hours.
Customer managed
You manage the storage for your query results and keep the results as long as you require.
Encrypt query results
Encrypt using AWS owned key (Default)
A key that AWS owns and manages for you.
Encrypt using customer managed key
Choose a key you have permission to use, or create a new one.
Override client-side settings Info
If checked, workgroup settings apply to all queries in the workgroup.

Manage query result location and encryption

Location of query result - *optional*

Enter an S3 prefix in the current region where the query result will be saved as an object.

s3://healthcare-1368762273 | View | Browse S3

You can create and manage lifecycle rules for this bucket

Use Amazon S3 lifecycle rules to store your query results and metadata cost effectively or to delete them after a period of time. Learn more

Lifecycle configuration

Expected bucket owner - *optional*

Specify the AWS account ID that you expect to be the owner of your query results output location bucket.

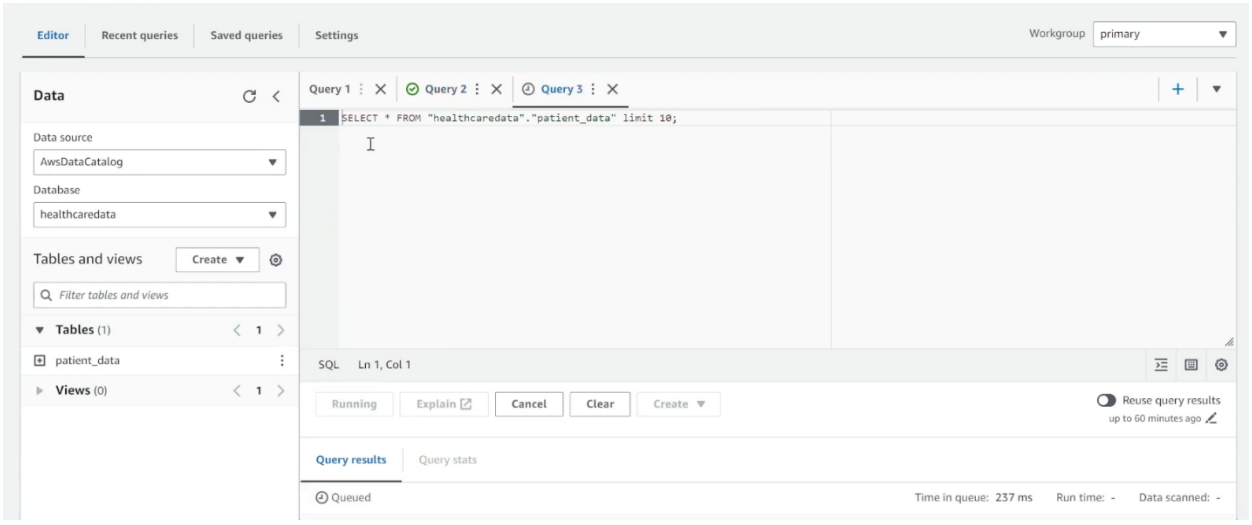
Enter AWS account ID

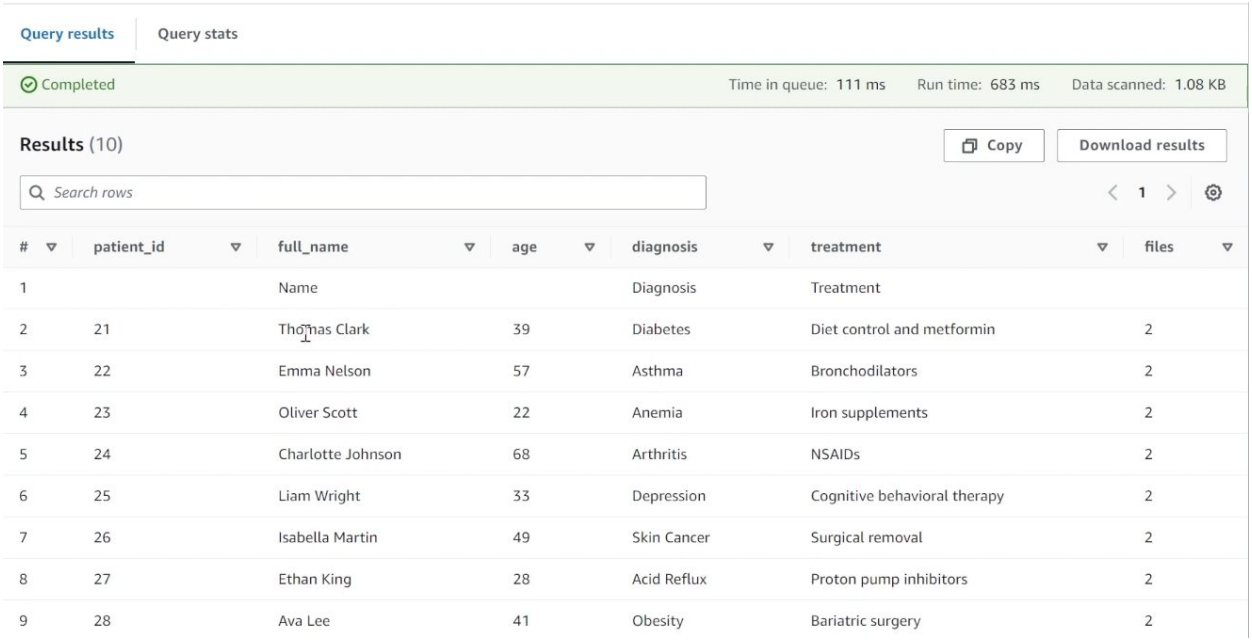
☐ Assign bucket owner full control over query results

Enabling this option grants the owner of the S3 query results bucket full control over the query results. This means that if your query result location is owned by another account, you grant full control over your query results to the other account.

☐ Encrypt query results

Cancel | Save





Query results | Query stats

Completed — Time in queue: 111 ms — Run time: 683 ms — Data scanned: 1.08 KB

Results (10)

Copy | Download results

Search rows

1

#	patient_id	full_name	age	diagnosis	treatment	files
1		Name		Diagnosis	Treatment	
2	21	Thomas Clark	39	Diabetes	Diet control and metformin	2
3	22	Emma Nelson	57	Asthma	Bronchodilators	2
4	23	Oliver Scott	22	Anemia	Iron supplements	2
5	24	Charlotte Johnson	68	Arthritis	NSAIDs	2
6	25	Liam Wright	33	Depression	Cognitive behavioral therapy	2
7	26	Isabella Martin	49	Skin Cancer	Surgical removal	2
8	27	Ethan King	28	Acid Reflux	Proton pump inhibitors	2
9	28	Ava Lee	41	Obesity	Bariatric surgery	2